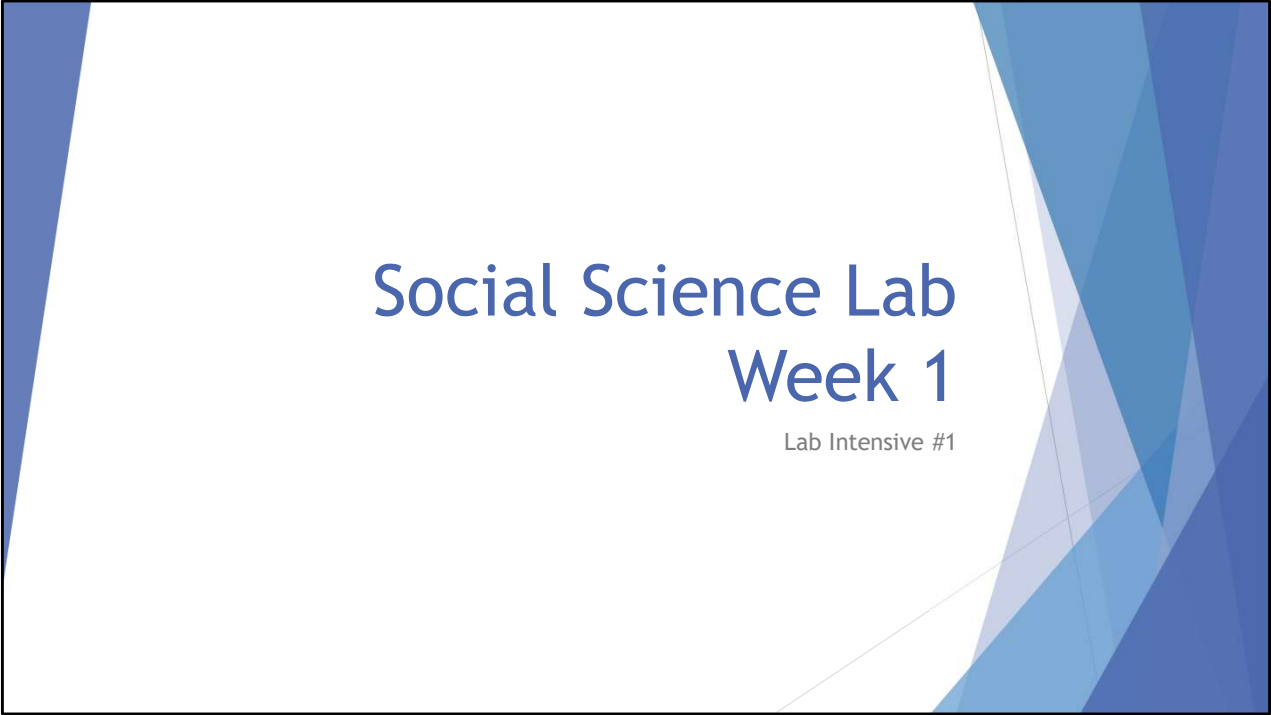
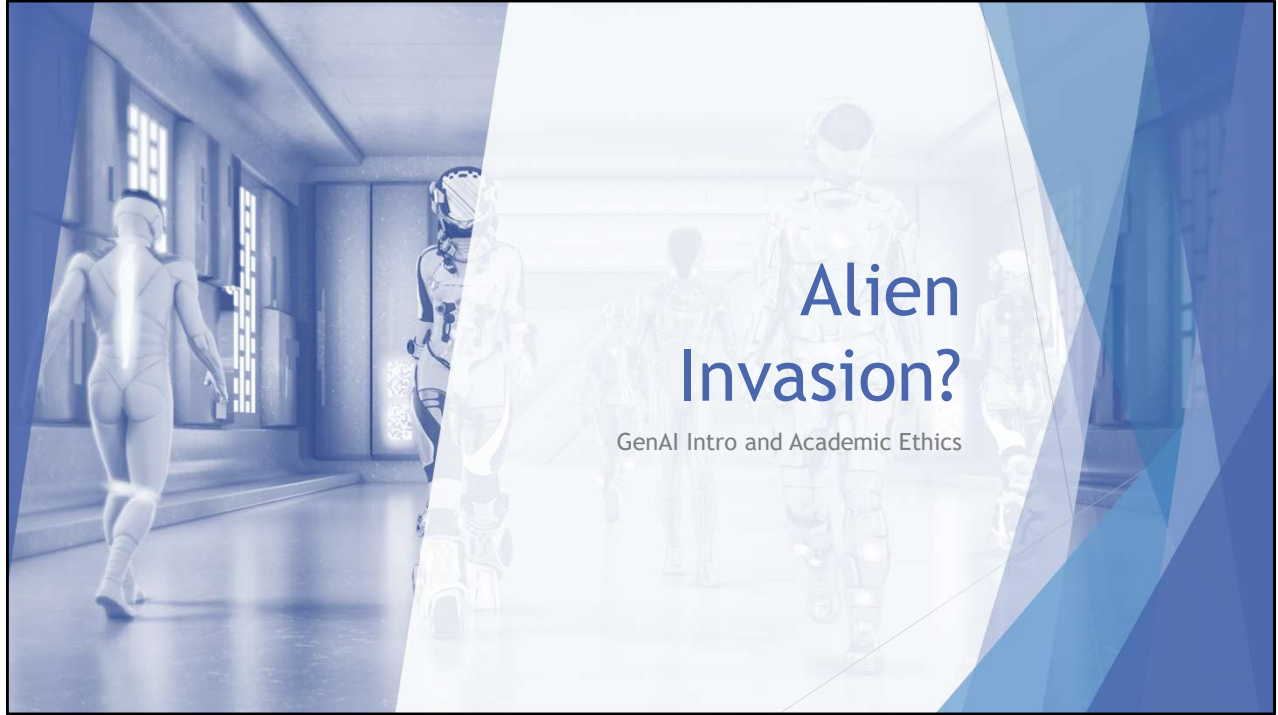




Social Science Lab Week 1

Lab Intensive #1



An Alien Encounter Event



In early 2025, Earth receives a signal from a small fleet of alien probes that have entered orbit. The probes broadcast a simple message in multiple languages:



“We come not to rule, not to save, but to study. Our species seeks to understand how humans make sense of their world and predict their own future. We have chosen not to speak to governments, armies, or corporations — we are sending our signals directly to classrooms.”



The probes appear benign — they hover silently, no weapons, no threats, just observation. Their intent: to collect human analysis of world affairs.

Why This Class?

One probe's signal lands directly at our college campus and connects to this IR course.

The aliens explain:

- ▶ They want non-expert but educated analysts – people who are still learning but can think critically.
- ▶ Experts may be biased; governments may be secretive. Students, they argue, are a better test of humanity's honest reasoning.
- ▶ This IR class has been “randomly” chosen (or so the aliens say) to present their research in written reports before the end of the semester.



**ORGANIZED
REPORTS AND
ANALYSIS FOR
CONTACT WITH
LIFE
ELSEWHERE**

Project O.R.A.C.L.E.

Over the next **six weeks**, we will explore international relations concepts through the lens of explaining humanity to these new observers. Each week's lab will ask:

- *How can IR help make sense of human behavior globally?*
- *How can expert research methods be useful for our reporting?*
- *What would aliens learn about us if they read our analyses?*
- *How could these tools help predict what we might do in the future — in peace, in conflict, in cooperation?*

We are not diplomats, generals, or presidents. We are **Project ORACLE — the eyes and minds that will help others make sense of Earth.**

Lab Goals (why are we doing this?)

1. To **give students hands-on exposure to research methods** used to answer contemporary IR questions and challenges.
2. To **foster AI literacy and ethics** in students.
3. To **develop marketable skills for students** by pairing Generative AI (GenAI) with practical research methods and academic standards.
4. To **foster interest and excitement for Political Science** through coursework based on current issues, student-driven sub-topics, and engaging technology.
5. To extend assignment mechanisms in the field of political science **beyond traditional academic research and writing skills**.
6. To **exemplify higher-level AI tool/skills** through workforce-ready tasks.
7. To **enhance students' awareness of their marketable skills** beyond the lab and classroom environment.

Lab Objectives (what will we do?)

1. Students will be able to **describe** the basic functions of general large language models (LLM) and **express** their limitations in education in 2025.
2. Students will be able to **describe** and **differentiate** academic, personal, and professional Generative AI-use ethics.
3. Students will **explore** "advanced" AI tools and prompting techniques regularly during instructor-created-and-led labs.
4. Students will **develop** their own "advanced" AI app to solve a practical question from contemporary IR studies.
5. Students will **reflect** on the skills developed and **identify** those most likely to improve their workforce readiness.

Lab Intensive #1 Session #1

What's IR say about alien encounters?

Alien Encounters: What's IR have to say?

Topic: This session explores how International Relations (IR) can explain and predict scenarios, using an "alien encounter" as a case study.

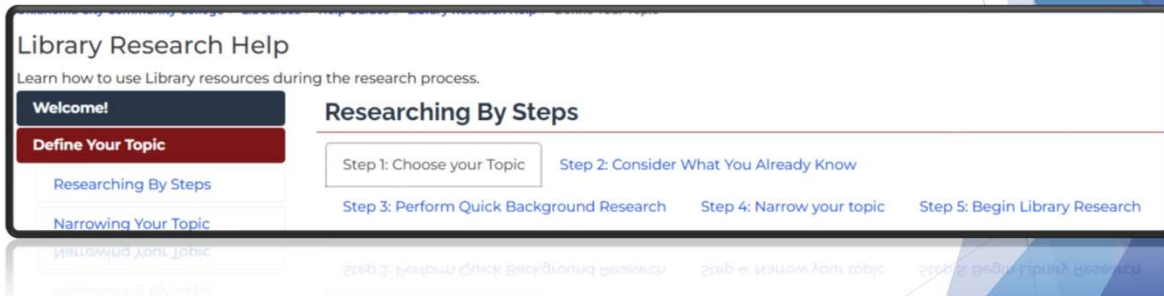
Takeaway: The key takeaway is understanding how to use critical thinking and Generative AI (Gen AI) to extract information from expert sources when research interests do not directly align with common search terms, all while maintaining academic integrity.

What does IR have to say about alien encounters?

- ▶ First, consider our **textbook** - our primary source and valid for most fundamental education, but not without potential mistrust.
- ▶ How would you find what the textbook says about alien encounters?
- ▶ How do we balance textbook's reliability with its limited coverage?

How about more academic resources here at OCCC?

- ▶ OCCC library has a “[Researching by Steps](https://libguides.occc.edu/research/topic)” guide that can be helpful



Source: <https://libguides.occc.edu/research/topic>

Let's “brute
force” it by
defining and
refining...

OCCC
EBSCOHost

OCCC EBSCO link source:

<https://occc.idm.oclc.org/login?url=http://search.ebscohost.com/login.asp?profile=ehost>

Exploring a topic can help you two ways—you can narrow your topic and also see if you can locate enough information.
Your focused topic and keywords will then help you find relevant credible sources.

What

- ...is it?
- ...happened?
- ...is the issue?
- ...are the synonyms?

Concept Cloud

Write down anything and everything related to your main idea.
(Use Google or Wikipedia to help here.)

The main idea you want to research.

Who

- ...is impacted?
- ...does it matter to?
- ...supports it?
- ...doesn't support it?

How and Why

- ...does it impact you or others?
- ...did it/does it happen?
- ...does it matter?

When and Where

- ...did it/does it happen?
- ...did it/does it make an impact?

Keywords: Select two to four keywords to combine together from the box above. You will use these keyword combinations to search for credible sources.

Remember: Add or subtract keywords to fine tune your search results.

Adapted from University of Virginia Library's "Thinking Tool: Choosing a Topic and Search Terms" by Burks and Wolnick

Source: https://libguides.occc.edu/ld.php?content_id=71762807

What about Generative Artificial Intelligence?

USER EXPERIENCE QUOTES

"This exercise has **helped me be more open to using AI**. I was personally against using it because I was afraid of my work being seen as plagiarism. With this assignment I have learned how to use it and how **helpful it really** is. It helped with understanding at a beginner level."

"**GenAI** explained this **step to me in more common**, simpler terms, such as keywords like **tariffs**, **rules**, and policies that are different throughout each country around the world.

Applying **STEEP to GenAI...** gave me **more lay terms** and more understandable examples relating to each topic under steep.

When using GenAI I felt like **I got a way better explanation...** I was not understanding when I asked the model in class.

The examples I was given were very **straightforward** and easy for me to understand. I personally need an **example** of anything I am learning to fully understand it, so this helped me a lot because I was very confused in my original attempt.

ChatGPT helps me... **because it isn't using words** that I don't understand.

It made **the topic** feel less like a **broad theory**, and at least more like a tool for understanding specific problems in technology or the environment.

help find the best way I could open my eyes and showed me a part of the deconstruction of our realities that I had never considered before

What are some
good use-cases
for GenAI at this
step in our task?

Let's brainstorm
some ideas

Why does this work so well?

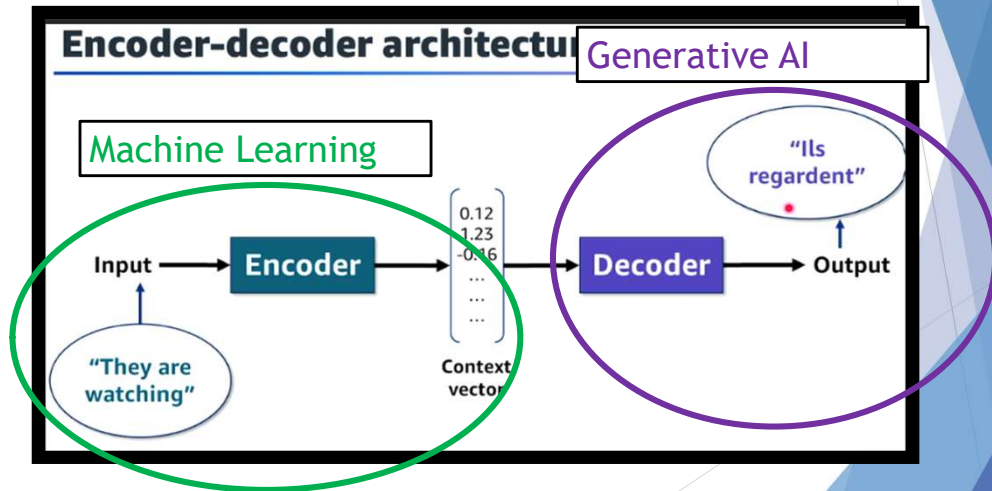
Generative Artificial Intelligence (GenAI) like Grok, Claude, GPT, etc. are based on Large Language Models (LLMs).

- ▶ What is a GenAI actually generating?
 - ▶ Text: The most likely (average) next word *based on context*
 - ▶ Images: The most likely (average) next pixel *based on context*

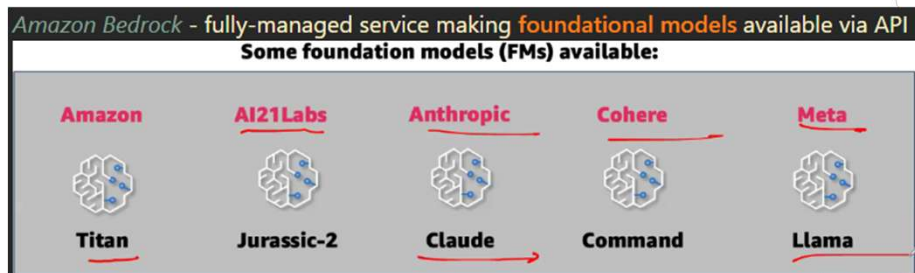
Some examples...

- ▶ We hold these truths to be self-evident that all men are _____
- ▶ With great power comes great _____
- ▶ I'm not giving away my _____
- ▶ Many science curricula do not ask even _____
- ▶ Pink pony _____
- ▶ Top _____
- ▶ If _____
- ▶ The _____

What's going on "under the hood" of large language models?



Different companies use distinct
“foundational models” of LLMs for
their GenAI bots



How to make use of LLMs ethically

- ▶ Academic ethics
- ▶ Personal ethics
- ▶ Professional ethics

How to make use of LLMs ethically: in academics

College Policy No. 5076 Student Community Guidelines

This policy applies to all students and student clubs and organizations on campus, as well as those participating in off-campus OCCC activities, classes, programs, or events. Student Clubs and Organization are also responsible for adhering to the current Student Club and Organization manual. The following misconduct is prohibited and is subject to disciplinary action.

2. Academic Misconduct: Cheating, plagiarism, unauthorized collaboration, alteration of academic materials or other academic misbehavior as specified in the Academic Integrity policy (policy 4016).

College policy link: <https://www.occc.edu/wp-content/uploads/5076-Student-Community-Guidelines.pdf>